

DUET C / C++ Program Reference

1. Fibonacci Series

```
C
#include <stdio.h>
int main() {
    int n, a = 0, b = 1, next;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
        printf("%d ", a);
        next = a + b;
        a = b;
        b = next;
    }
    return 0;
}
```

```
C++
#include <iostream>
using namespace std;
int main() {
    int n, a = 0, b = 1, next;
    cout << "Enter number of terms: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        cout << a << " ";
        next = a + b;
        a = b;
        b = next;
    }
    return 0;
}
```

2. GCD and LCM

```
C
#include <stdio.h>
int main() {
    int a, b, x, y, gcd, lcm;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    x = a; y = b;
    while (y != 0) {
        int temp = y;
        y = x % y;
        x = temp;
    }
    gcd = x;
    lcm = (a * b) / gcd;
    printf("GCD = %d\n", gcd);
    printf("LCM = %d\n", lcm);
    return 0;
}
```

```
C++
#include <iostream>
using namespace std;
int main() {
    int a, b, x, y, gcd, lcm;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    x = a; y = b;
    while (y != 0) {
        int temp = y;
        y = x % y;
        x = temp;
    }
    gcd = x;
    lcm = (a * b) / gcd;
    cout << "GCD = " << gcd << endl;
    cout << "LCM = " << lcm << endl;
    return 0;
}
```

3. Sum of Digits

```
C
#include <stdio.h>
int main() {
    int num, temp, r, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum + r;
        temp = temp / 10;
    }
    printf("Sum of digits = %d\n", sum);
    return 0;
}
```

```
C++
#include <iostream>
using namespace std;
int main() {
    int num, temp, r, sum = 0;
    cout << "Enter a number: ";
    cin >> num;
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum + r;
        temp = temp / 10;
    }
    cout << "Sum of digits = " << sum << endl;
    return 0;
}
```

4. Reverse Number

```
C
#include <stdio.h>
int main() {
    int num, temp, r, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum * 10 + r;
        temp = temp / 10;
    }
    printf("Reversed number = %d\n", sum);
    return 0;
}
```

```
C++
#include <iostream>
using namespace std;
int main() {
    int num, temp, r, sum = 0;
    cout << "Enter a number: ";
    cin >> num;
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum * 10 + r;
        temp = temp / 10;
    }
    cout << "Reversed number = " << sum << endl;
    return 0;
}
```

5. Palindrome Check

```
C
#include <stdio.h>
int main() {
    int num, temp, r, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum * 10 + r;
        temp = temp / 10;
    }
    printf(sum == num ? "Palindrome number\n" : "Not
    palindrome number\n");
    return 0;
}
```

```
C++
#include <iostream>
using namespace std;
int main() {
    int num, temp, r, sum = 0;
    cout << "Enter a number: ";
    cin >> num;
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum * 10 + r;
        temp = temp / 10;
    }
    cout << (sum == num ? "Palindrome number\n" : "Not
    palindrome number\n");
    return 0;
}
```

6. Armstrong Number (3-digit)

```
C
#include <stdio.h>
int main() {
    int num, temp, r, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum + r * r * r;
        temp = temp / 10;
    }
    printf(sum == num ? "Armstrong number\n" : "Not
    Armstrong number\n");
    return 0;
}
```

```
C++
#include <iostream>
using namespace std;
int main() {
    int num, temp, r, sum = 0;
    cout << "Enter a number: ";
    cin >> num;
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum + r * r * r;
        temp = temp / 10;
    }
    cout << (sum == num ? "Armstrong number\n" : "Not
    Armstrong number\n");
    return 0;
}
```

7. Strong Number

```
C
#include <stdio.h>
int main() {
    int num, temp, r, sum = 0, f;
    printf("Enter a number: ");
    scanf("%d", &num);
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        f = 1;
        for (int i = 1; i <= r; i++)
            f = f * i;
        sum = sum + f;
        temp = temp / 10;
    }
    printf(sum == num ? "Strong number\n" : "Not strong
    number\n");
    return 0;
}
```

```
C++
#include <iostream>
using namespace std;
int main() {
    int num, temp, r, sum = 0, f;
    cout << "Enter a number: ";
    cin >> num;
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        f = 1;
        for (int i = 1; i <= r; i++)
            f = f * i;
        sum = sum + f;
        temp = temp / 10;
    }
    cout << (sum == num ? "Strong number\n" : "Not
    strong number\n");
    return 0;
}
```

8. Armstrong Number (Any Digit)

```
C
#include <stdio.h>
#include <math.h>
int main() {
    int num, temp, r, sum = 0, count = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    temp = num;
    while (temp != 0) {
        count++;
        temp = temp / 10;
    }
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum + pow(r, count);
        temp = temp / 10;
    }
    printf(sum == num ? "Armstrong number\n" : "Not
Armstrong number\n");
    return 0;
}
```

```
C++
#include <iostream>
#include <cmath>
using namespace std;
int main() {
    int num, temp, r, sum = 0, count = 0;
    cout << "Enter a number: ";
    cin >> num;
    temp = num;
    while (temp != 0) {
        count++;
        temp = temp / 10;
    }
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum + pow(r, count);
        temp = temp / 10;
    }
    cout << (sum == num ? "Armstrong number\n" : "Not
Armstrong number\n");
    return 0;
}
```